

Portfolio Temporal Structure Learning with Dynamic Bayesian Networks: Learning Causal Asset Dependencies from Time-Series Data

Category: Applied AI

Group Name: G7_APL_Finance_team

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1. Problem Statement

Financial time-series exhibit strong temporal dependencies: asset returns today influence returns tomorrow, and these dependencies vary across market regimes. Traditional econometric models (VAR, GARCH) capture temporal correlation but do not explicitly learn the *structure* of which assets directly predict which others. This project addresses structure learning in portfolio time-series: given daily returns of multiple assets, learn a directed acyclic graph (DAG) encoding temporal causal relationships (e.g., does Apple price predict Microsoft price tomorrow?). Understanding these structures is valuable for:

- Portfolio construction: identify assets that lead vs. follow
- Risk management: detect contagion pathways in time-lagged dependencies
- Interpretability: explicit edge structure reveals direct relationships, not just correlation

2. Project Summary

We apply dynamic Bayesian network (DBN) structure learning to portfolio asset returns. Dynamic Bayesian networks are directed graphical models that explicitly model temporal dependencies: nodes are assets, edges represent conditional dependence at time t or $t - \tau$, and acyclicity ensures causal consistency. We adopt the score-based learning framework of DYNOTEARS [1], which learns both contemporaneous and lagged edges by optimizing a penalized score subject to an acyclicity constraint. Our approach: (i) load daily returns for 10–15 portfolio assets, (ii) fit linear autoregressive models to learn which assets predict others at lag-1, (iii) extract edge structure from regression coefficients and statistical significance, (iv) validate learned DAGs using out-of-sample prediction accuracy. The comparison baseline is vector autoregressive (VAR) models, which estimate full coefficient matrices without structure learning. Our contribution is demonstrating that score-based structure learning recovers interpretable causal structures in real portfolio data.

3. Representation: Dynamic Bayesian Network

A dynamic Bayesian network is a directed graphical model $G = (V, E)$ where:

- Nodes $V = \{X_1^t, \dots, X_n^t\}$: assets at time slices t and $t - 1$
- Edges E : directed edges $X_i^{t-1} \rightarrow X_j^t$ (lag-1) or $X_i^t \rightarrow X_j^t$ (contemporaneous)
- Acyclicity constraint: edges within a time slice form a DAG; lagged edges always point forward in time

For portfolio returns $\mathbf{R}_t = (R_1^t, \dots, R_n^t)$, the joint distribution factorizes as:

$$P(\mathbf{R}^1, \dots, \mathbf{R}^T) = P(\mathbf{R}^1) \prod_{t=2}^T P(\mathbf{R}^t | \mathbf{R}^{t-1}, \mathbf{R}_{pred}^t)$$

where $\mathbf{R}_{pred}^t \subseteq \mathbf{R}^t$ are the parent nodes of $\mathbf{R}^t$ in the contemporaneous slice.

4. Learning: Score-Based Structure Learning

We learn the DAG structure by minimizing a penalized score over all acyclic directed graphs:

$$\hat{G} = \arg \min_{G \in \mathcal{G}} [Score(G; \mathbf{R}) + \lambda |E|]$$

where:

- $Score(G; \mathbf{R})$ = BIC (Bayesian Information Criterion): goodness-of-fit minus model complexity
- $\lambda |E|$ = sparsity penalty (number of edges)
- $\mathcal{G}$ = set of all acyclic directed graphs

Acyclicity is enforced via a *topological ordering constraint*: if graph is acyclic, there exists a permutation of nodes such that all edges point in increasing order. We search over orderings and fit linear models for each ordering.

M1 Simplification: For proof-of-concept, we fit lag-1 autoregressive models (ARX) independently per asset, extract edge structure from non-zero coefficients, and verify acyclicity. This avoids full optimization but demonstrates the learning concept.

5. Inference: Temporal Prediction and Edge Discovery

Given learned DAG structure, inference computes:

- Temporal prediction: $\hat{R}_j^t = \sum_{i \in \text{parents}(j)} \beta_{ij} R_i^{t-1} + \epsilon_j^t$
- Edge confidence: assess statistical significance of edge weights via t-tests
- Causal interpretation: directed edges imply *direct* temporal influence (not mediated by other assets)

6. State-of-the-Art Positioning

Base Paper: Pamfil et al., “DYNOTEARS: Structure Learning from Time-Series Data,” *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020.

Problem: Learn directed acyclic graphs (DAGs) from multivariate time-series data, simultaneously estimating contemporaneous and lagged edges.

Representation & Learning: DYNOTEARS formulates structure learning as score-based optimization with acyclicity constraint characterized algebraically (trace-based characterization). Uses second-order optimization (L-BFGS) to minimize BIC-penalized loss over DAGs.

Inference: Once structure is learned, predict future values given past observations and identify causal structures.

Results: Outperforms other DBN learning methods on simulated high-dimensional data. Applied to financial and biological datasets.

Limitations: Assumes acyclicity (no feedback loops in contemporaneous slice); assumes linear-Gaussian models; computational cost scales with network size.

Relationship to Our Project: We adopt DYNOTEARS’s framework (score-based learning + acyclicity) applied to portfolio returns. Our M1 work is a simplified proof-of-concept; subsequent milestones will implement full DYNOTEARS optimization.

7. Baseline & Key Performance Indicators (KPIs)

Baseline: Vector Autoregressive (VAR) model of order 1: fits full dense matrix of lagged coefficients without structure learning. We compare edge recovery and prediction accuracy.

KPIs:

- **Representation KPI:** Sparsity of learned DAG: edge count, density ($|E|/n^2$), mean degree
- **Learning KPI:** BIC score, coefficient magnitude distribution, number of significant edges ($p < 0.05$)
- **Inference KPI:** Out-of-sample prediction error (RMSE, MAE) on held-out test set; compare against VAR baseline
- **Structure KPI:** Edge recovery rate (if synthetic ground truth available); interpretability of discovered edges

8. Dataset

Primary: S&P 500 daily returns, 10–15 major stocks (AAPL, MSFT, GOOGL, AMZN, TSLA, JPM, GS, XOM, CVX, NVDA), 2015–2024 (2500 trading days). Downloaded via yfinance. Small enough for rapid iteration, large enough to capture market regimes.

Preprocessing: Compute log-returns $r_t = \log(p_t/p_{t-1})$. Standardize to zero mean, unit variance. Remove weekends/holidays. Check for stationarity (ADF test).

Train/Test Split: 80/20 temporal split (train on 2015–2022, test on 2023–2024).

9. Experimental Plan

M1 (Now):

- Load data, compute returns, standardize
- Fit lag-1 ARX models per asset; extract edge structure from significant coefficients
- Verify acyclicity via topological sorting
- Compute sparsity, BIC, prediction RMSE on test set

- Visualize learned DAG structure
- Compare against VAR baseline

M2 (Oct 11):

- Implement full score-based optimization (BIC + penalty)
- Test different regularization strengths (λ)
- Evaluate on multiple time periods and subsets

M3 (Nov 1):

- Full DYNOTEARS implementation or library wrapper
- Systematic comparison: VAR vs. structured learning
- Sensitivity analysis (lag-1 vs. lag-2, different λ)
- Interpret discovered edges in portfolio context

M4 (Nov 22):

- Final evaluation, limitations discussion
- Code release, reproducibility documentation

10. Team Responsibilities

- **Foram Gohel:** Data loading, preprocessing, VAR baseline implementation, prediction evaluation
- **Priya Iyengar:** Structure learning (ARX models, edge extraction), DAG visualization, KPI computation
- **Both:** Code documentation, README, video walkthrough, hypothesis interpretation

R. Pamfil et al., “DYNOTEARS: Structure Learning from Time-Series Data,” in *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020.

J. Friedman, T. Hastie, and R. Tibshirani, “Sparse inverse covariance estimation with the graphical lasso,” *Biostatistics*, vol. 9, no. 3, pp. 432–441, 2008.

D. Koller and N. Friedman, *Probabilistic Graphical Models: Principles and Techniques*. Cambridge, MA: MIT Press, 2009.

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References

- [1] R. Pamfil et al., “DYNOTEARS: Structure Learning from Time-Series Data,” in *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020.
- [2] J. Friedman, T. Hastie, and R. Tibshirani, “Sparse inverse covariance estimation with the graphical lasso,” *Biostatistics*, vol. 9, no. 3, pp. 432–441, 2008.
- [3] D. Koller and N. Friedman, *Probabilistic Graphical Models: Principles and Techniques*. Cambridge, MA: MIT Press, 2009.
- [4] H. Lütkepohl, *New Introduction to Multiple Time Series Analysis*. Springer, 2005.